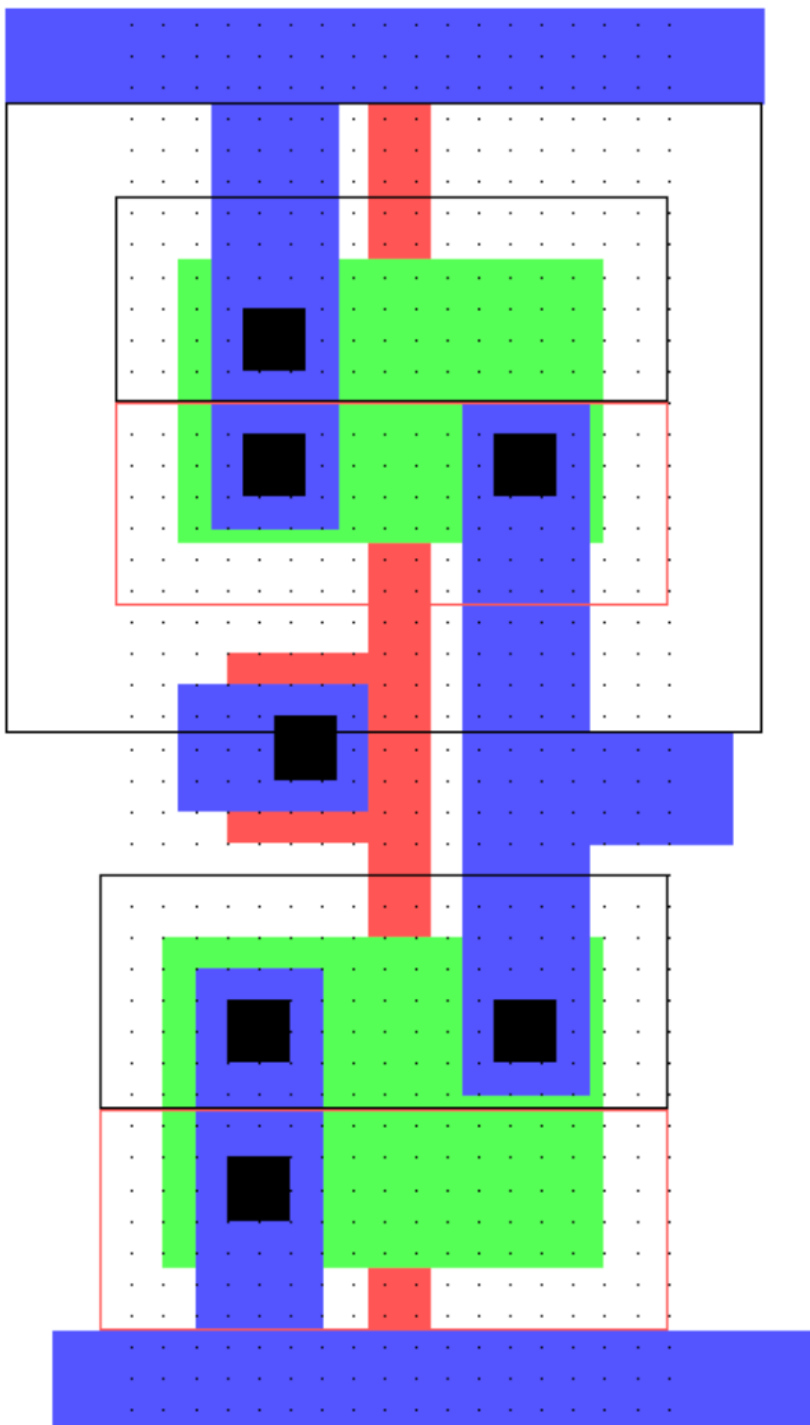
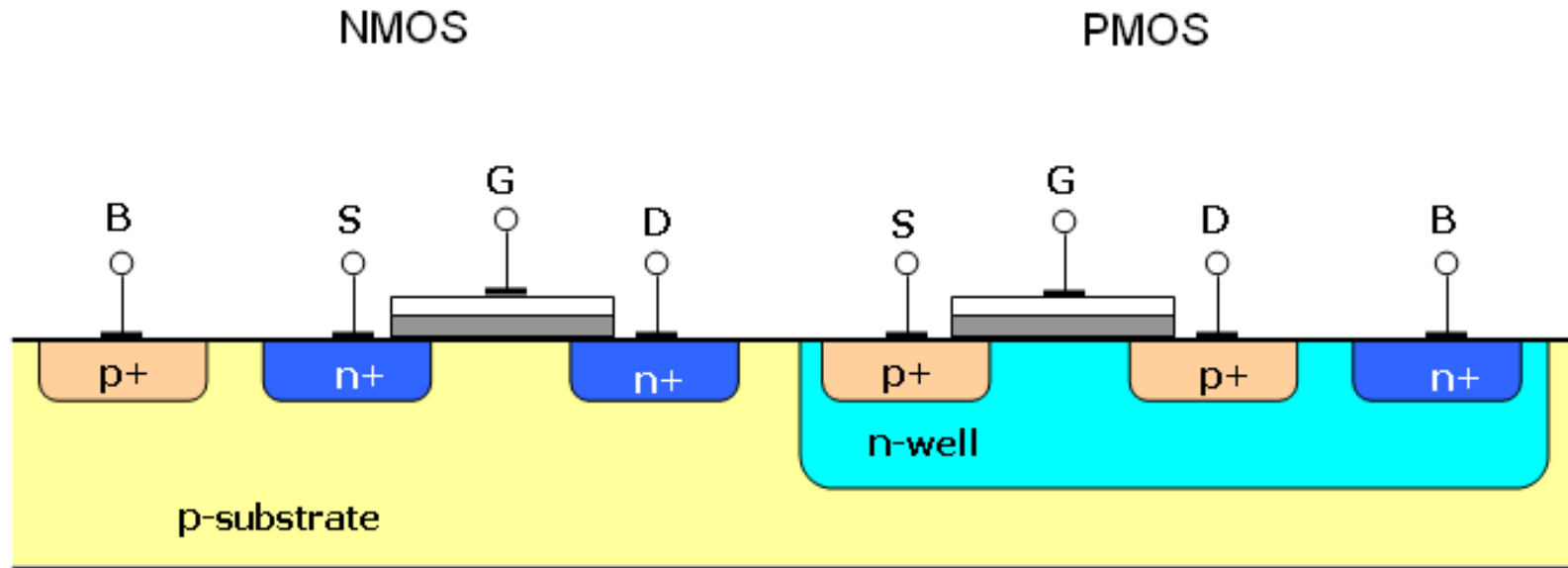




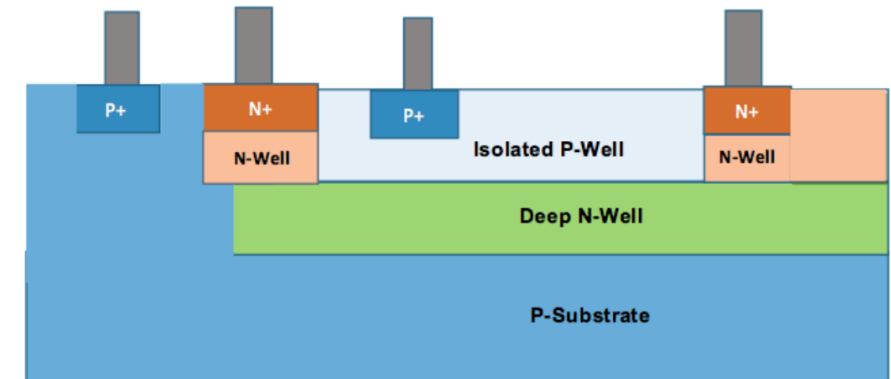
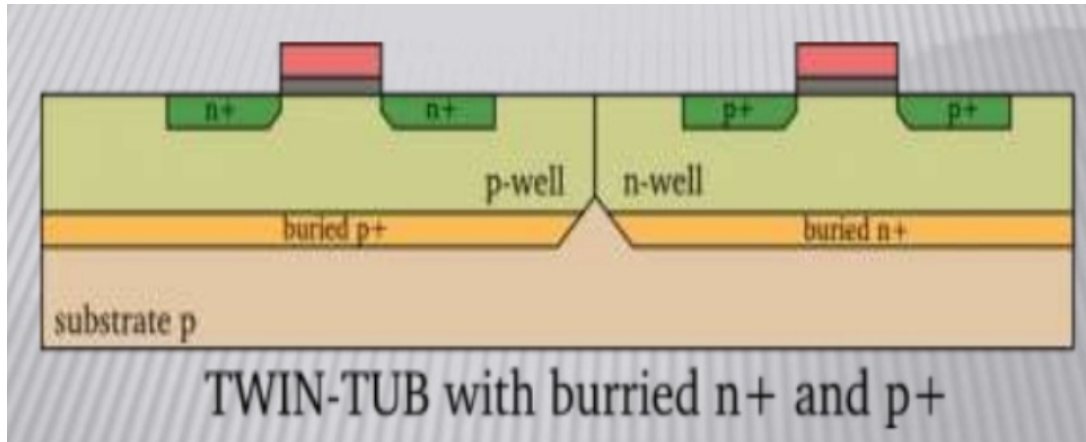
Richiamo alle Tecnologie Integrate

Tecnologia CMOS standard

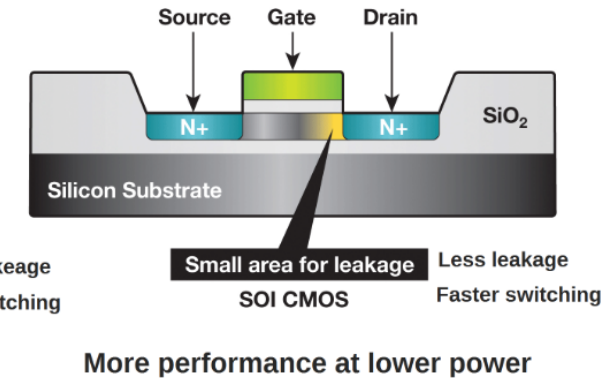
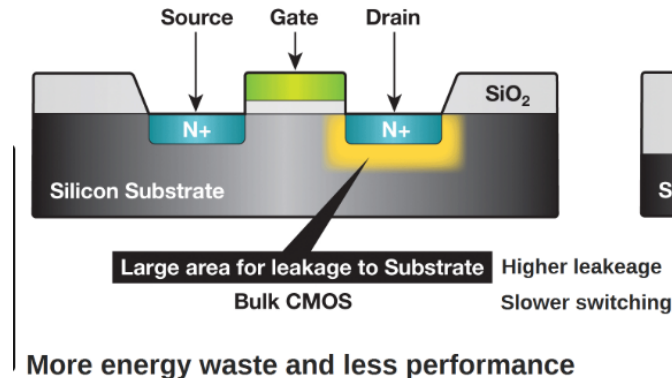


CMOS psub (detta anche nwell)

Altre tecnologie molto diffuse...



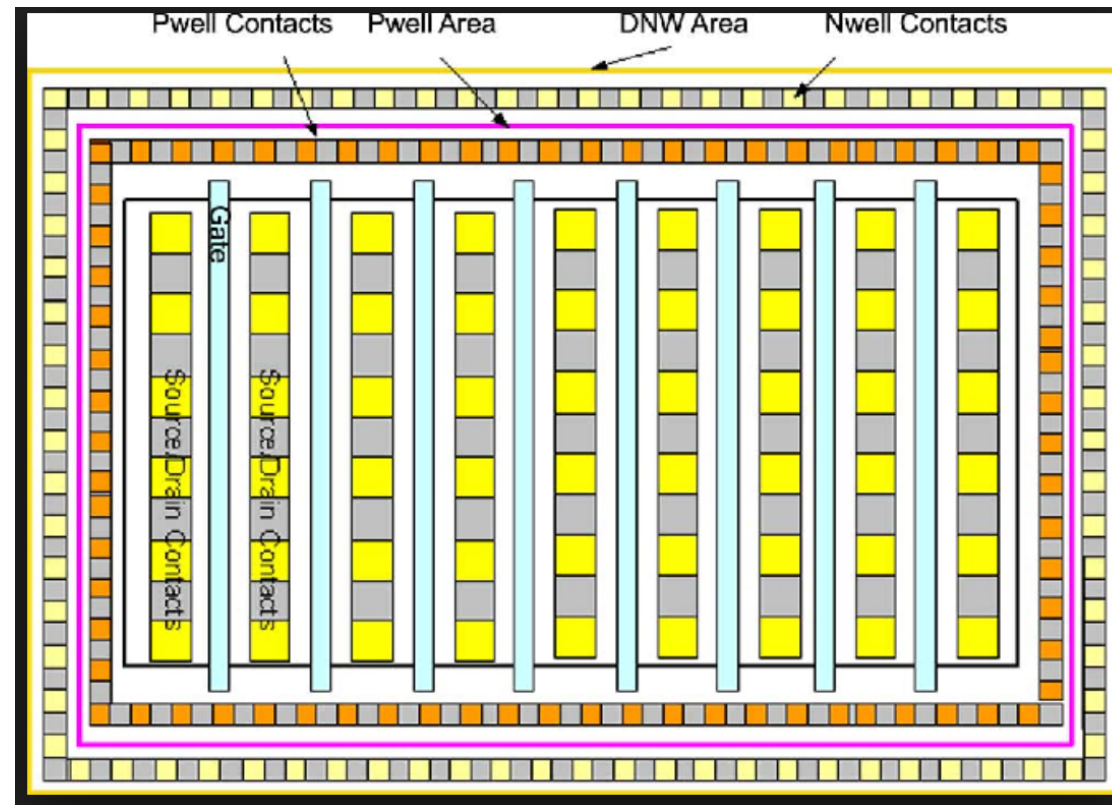
- CMOS twin-tub
- CMOS triple well
- SOI
- BiCMOS



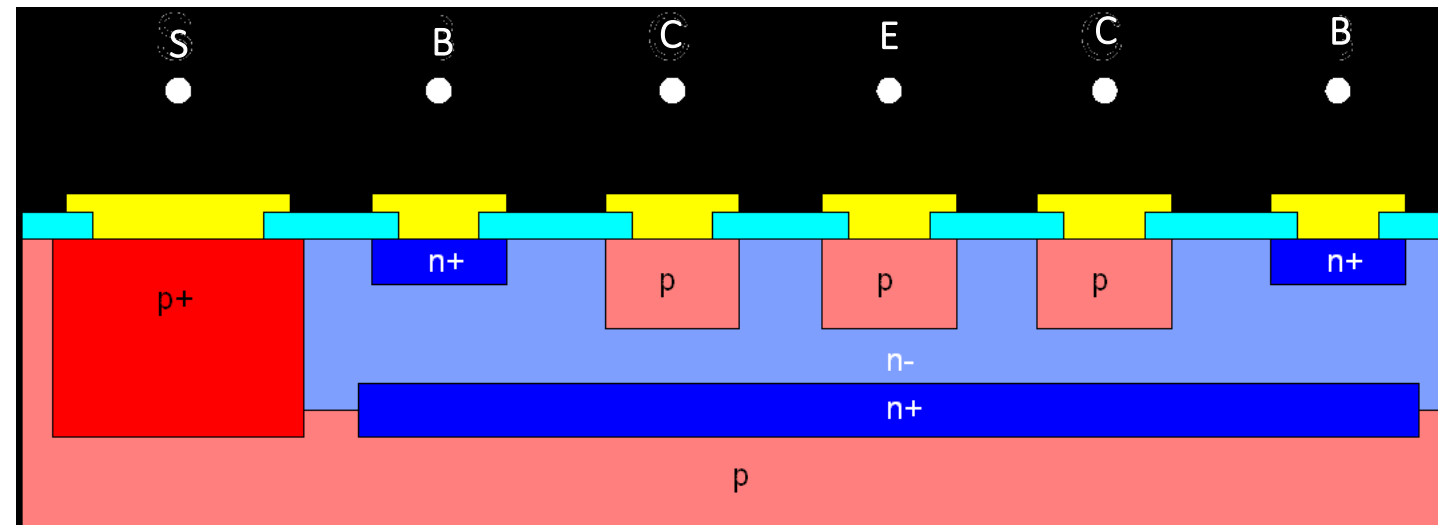
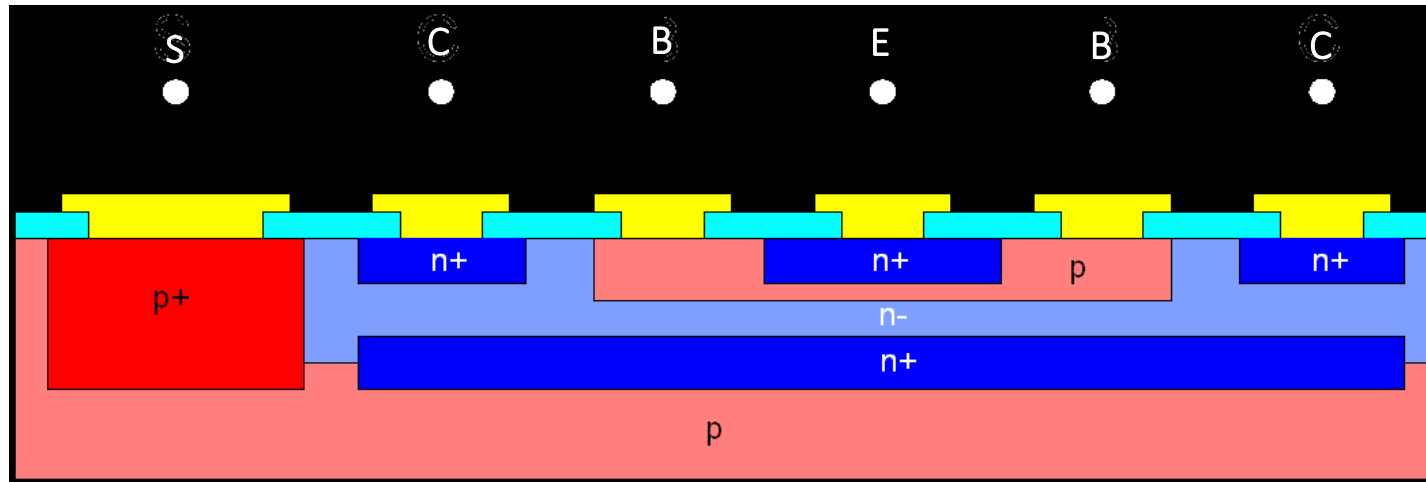
Alcune domande...

- Come si confrontano le tecnologie?
- Quale layout risulta più compatto?
- Quale meno compatto?
- Dal punto di vista del matching sono uguali?
- Quale tecnologia è più costosa?
- Quale tecnologia è migliore per applicazioni low power?

Che transistor è?



E questi, che roba sono?

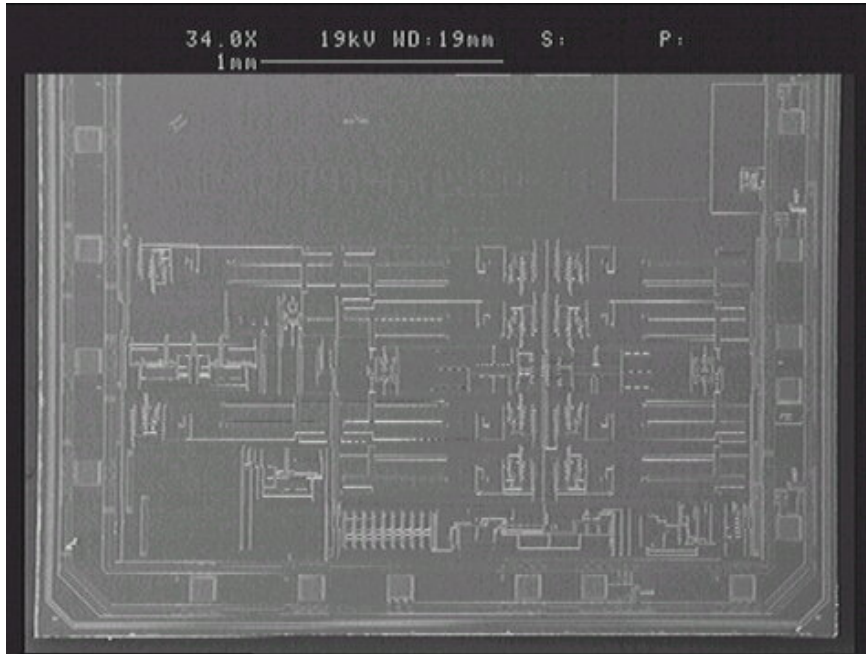


Se nei CMOS servono i BJT...

- In CMOS standard: substrate vertical pnp
- In BiCMOS: vertical npn, lateral pnp, substrate vertical pnp

Esempi di tecnologia: Alcatel

Fabbrica ← Tecnologia Alcatel 0.7 μ m
CMOS-A-2P2M n-well

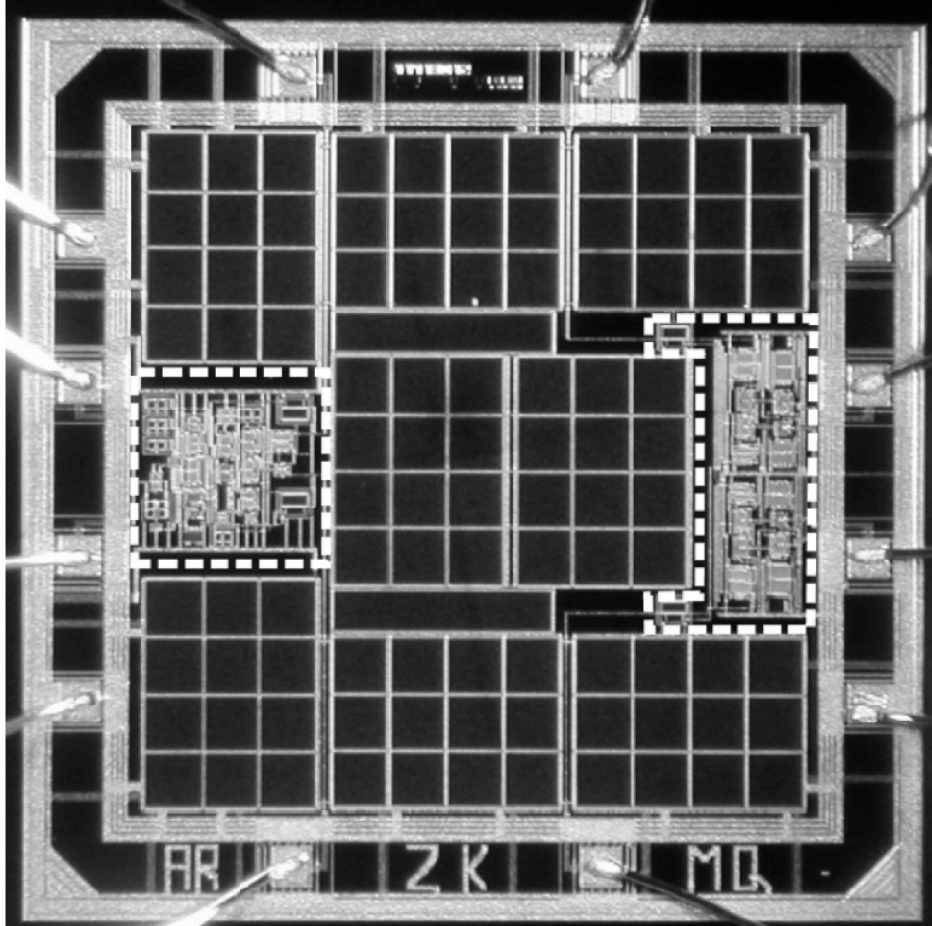


Lmin

Substrato p

CMOS analogica,
2 livelli di poli, 2 livelli di metal

Esempi di tecnologia: AustriaMicroSystems



Esempio di layout in
tecnologia AMS $0.8\mu\text{m}$

Esempi di tecnologia: AMS0.8 (i)

Process name	No. of masks	CMOS core module *	HV CMOS module	Poly1- poly2 capacitor module	High resistive poly module
CXT	15	x	x		
CXY	16	x	x	x	
CXZ	17	x	x	x	x

Esempi di tecnologia: AMS0.8 (ii)

Structural and Geometrical Parameters

PASS/FAIL PARAMETERS					
Parameter	Symbol	Min	Typ	Max	Unit
field oxide thickness	TFOX	350	420	490	nm
thin gate oxide thickness	TGOX	15.9	17.0	18.1	nm
POLY1 thickness	TPOLY1	270	300	330	nm

MOS 5V N-Channel Electr. Parameters: NMOS

These transistors are placed in the p-substrate.

PASS/FAIL PARAMETERS					
Parameter	Symbol	Min	Typ	Max	Unit
threshold voltage long channel 20x20	VTO20X20N	0.68	0.76	0.84	V
threshold voltage short channel 20x0.8	VTO20X08N	0.62	0.72	0.82	V

Esempi di tecnologia: AMS0.8 (iii)

POLY1-Resistor: RPOLY

PASS/FAIL PARAMETERS								
Parameter	Symbol	Min	Typ	Max	Unit		Unit	
POLY1 sheet resistance	RPOLY	27	32	37	$\Omega/\square$		$\Omega/\square$	
POLY1 effective width 0.8 μm	WPOLY	0.60	0.75	0.90	μm		μm	
INFORMATION PARAMETERS								
Parameter	Symbol	Min	Typ	Max	Unit		Unit	
POLY1 temperature coefficient	TCPOLY		0.7		$10^{-3}/\text{K}$		$10^{-3}/\text{K}$	

Sheet Resistances (continued)

PASS/FAIL PARAMETERS								
Parameter	Symbol	Min	Typ	Max	Unit		Unit	
NDIFF sheet resistance	RDIFFN	34	42	50	$\Omega/\square$		$\Omega/\square$	
PDIFF sheet resistance	RDIFFP	45	55	70	$\Omega/\square$		$\Omega/\square$	
MET1 sheet resistance	RMET	40	70	100	$\text{m}\Omega/\square$		$\text{m}\Omega/\square$	
MET2 sheet resistance	RMET2	20	35	60	$\text{m}\Omega/\square$		$\text{m}\Omega/\square$	

Esempi di tecnologia: AMS0.8 (iv)

Capacitances

INFORMATION PARAMETERS						
Parameter	Symbol	Min	Typ	Max	Unit	Note
MET2 – POLY1/POLY2						
area	CM2P	0.016	0.021	0.029	fF/ μm^2	29
Perimeter	CM2PF	0.035	0.038	0.042	fF/ μm	30
MET2 – MET1						
Area	CMM	0.026	0.038	0.069	fF/ μm^2	29
Perimeter	CMMF	0.041	0.047	0.056	fF/ μm	30
COUPLING CAPACITANCES						
POLY1 – POLY1	CP1P1		0.03		fF/ μm	33
MET1 – MET1	CM1M1		0.05		fF/ μm	33
MET2 – MET2	CM2M2		0.08		fF/ μm	33
POLY1 – MET1 – MET2 SANDWICH CAPACITANCE						
Area	CSANDWT	0.071	0.085	0.120	fF/ μm^2	29
perimeter	CSANDWTF	0.044	0.048	0.054	fF/ μm	30

Alcune domande sul matching...

- A cosa serve? Quando si fa?
- Ha degli svantaggi?
- Quali sono le regole di matching principali?
- Si può matchare una coppia differenziale bulk-driven?
- Come si simula/determina l'importanza del matching?

Le regole di matching funzionano?

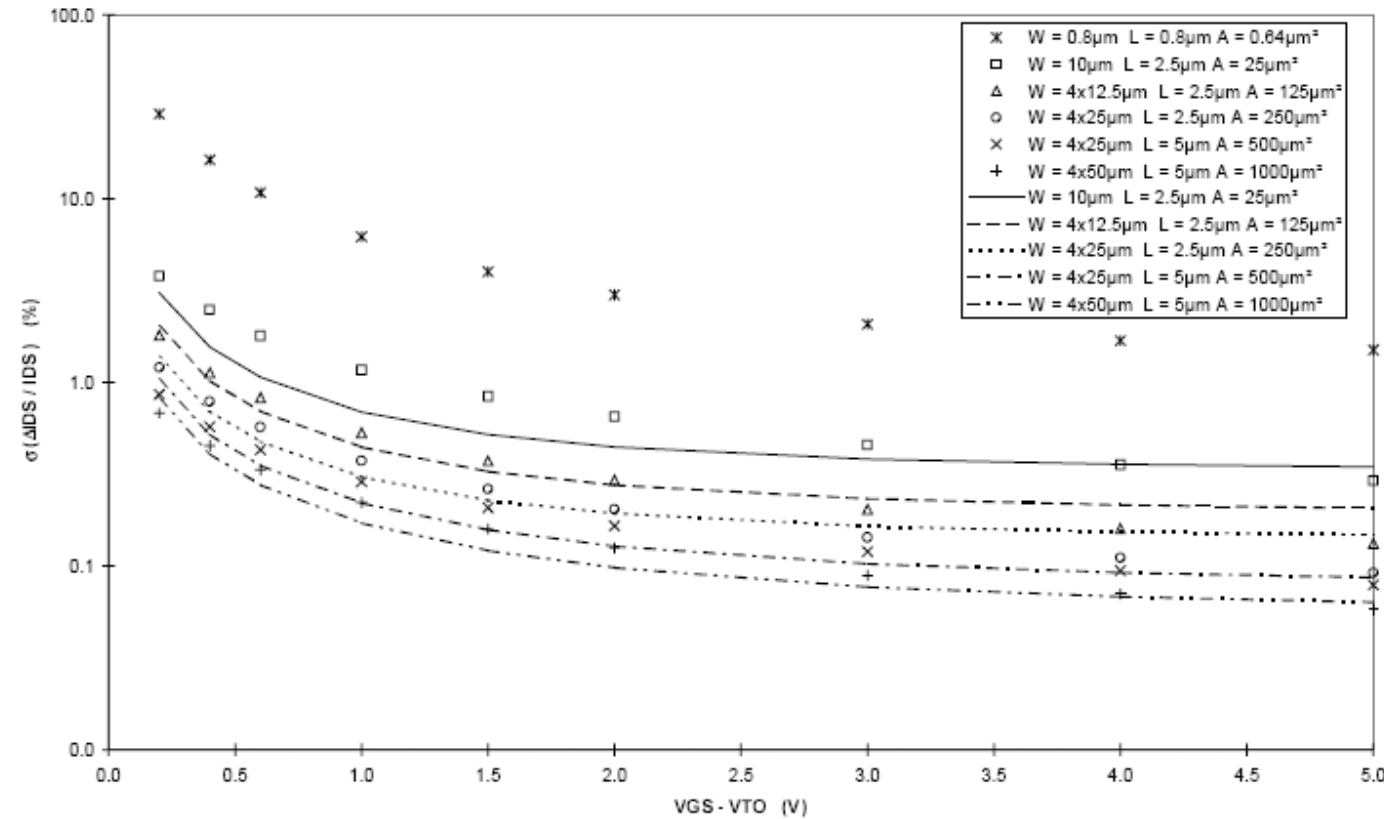


Fig. 4.1 NMOS drain current matching vs V_{GS} (typical values).
symbols = measured data, lines = simulation (currently simulation curves exist
for non-minimum transistors only)

La nostra tecnologia: UMC 0.18 μ m

- Tecnologia Europractice
- Fabbrica a Taiwan
- 1 Poly, 6 Metal
- Transistori per $V_{dd}=1.8V$ e $V_{dd}=3.3V$
- V_T standard, bassa e zero V_T (solo NMOS)

Vediamo di conoscerla!

Chi progetta «usa» I_B , W/L , L e ΔV

Prendiamo i transistor della libreria:
i 18_MM, i 33_MM e gli LV

Li connettiamo a diodo e li polarizziamo con
un generatore ideale di corrente da $1\mu A$, $10\mu A$ e $100\mu A$.

Consideriamo tre casi:

W e L minimi, $L=2 * L_{min}$ e $W=10 * W_{min}$,
 $L=10 * L_{min}$ e $W=10 * W_{min}$

In una tabella riportiamo g_m , g_{ds} , V_T e ΔV (simulazione DC, .OP)

